MILITARY SPECIFICATION

FUZE, IMPACT: PYROTECHNIC, XM733

METAL PARTS FOR

AND

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the metal parts for and loading, assembling and packing for one type of fuze designated as Fuze, Impact, Pyrotechnic, XM733.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1390

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MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-1168 - Lot Numbering of Ammunition

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS
U.S. Army Munitions Command

9277712 - Fuze, Impact: Pyrotechnic, XM733

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transporatation 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)".)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Assembly.-The assembly shall comply with all requirements specified on Drawing (Dwg.) 9277712, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

3.4 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (see 6.4).

3.5 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the follow-

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ing information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements when such material is purchased by the contractor or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1.) The first article sample shall consist of the following:

a. Ten (10) casing after forming.
b. Ten (10) supports.
c. Ten (10) springs.
d. Twenty (20) casing assemblies.
e. Twenty (20) delay assemblies.
f. Twenty (20) fuze assemblies.

All parts and completed assemblies submitted shall have been produced by the Contractor (or furnished by a supplier) using the same production processes, procedures and equipment as

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will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process (including heat treatment procedures and equipment, material used, drawing specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspection to be performed.-All parts and assemblies shall be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawing.

4.2.3 Rejection.-If any part or assembly fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any part or assembly. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.-

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with

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applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Change to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Heat mix from one interfix lot number from one manufacturer.
b. Transfer mix from one interfix lot number from one manufacturer.
c. Output mix from one interfix lot number from one manufacturer.
d. First fire mix from one interfix lot number from one manufacturer.
e. Time delay mix from one interfix lot number from one manufacturer.
f. Parts from one lot number from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plan for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Individual defects shall have an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Casing after forming (See Dwg. 9281579).-

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical: None defined			
Major: None defined			
Minor			
201.	Concentricity of flange with tube diameter	Gage	01001
202.	Perpendicularity	Gage	01002
203.	Evidence of poor workmanship (see 3.3)	Visual	01003

4.3.2.2 Casing assembly (See Dwg. 9281578).-

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major			
101.	Casing cracked	Visual	02001
Minor			
201.	Roll over of casing not uniform	Visual	02002

4.3.2.3 Support (See Dwg. 9281576).-

Categories	Defects	Method of Inspection (See 4.3.3.1)	Code No.
Critical: None defined			
Major: None defined			
Minor			
201.	Spring hole diameter	Gage	03001
202.	Total length	Gage	03002

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203.	Perpendicularity	Gage	03003
204.	Evidence of poor workmanship (see 3.3)	Visual	03004

4.3.2.4 Delay assembly (See Dwg. 9281573).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Cylinder missing	Visual	04001
102.	Excess adhesive internally or externally	Visual	04002
Minor			
201.	Evidence of poor workmanship (See 3.3)	Visual	04003

4.3.2.5 Spring (See Dwg. 9281577).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor			
201.	Fuze length	Gage	05001

4.3.2.6 Fuze assembly (See Dwg. 9277712).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Total length	Gage	06001
102.	Tape missing over spring hole	Visual	06002
103.	Tube not cemented or is excessively distorted	Visual	06003

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Minor			
201.	Solution missing at required length	Visual	06004
202.	Retainer not secure	Visual	06005
203.	Solution not support	Visual	06006
204.	Evidence of poor workmanship (see 3.3)	Visual	06007

4.3.3. Testing.-

4.3.3.1 Dimensional control of plastic parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, and after a stabilization time for the parts has been approved (see Note 1), a sample of five (5) parts (as molded) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of each cavity during production.

The inspection of the parts shall be performed only when the stabilization time for the parts has been reached. The inspection for control of each cavity during production shall be performed on the above quantities from each cavity at the beginning of production and at intervals of 5,000 parts, or at the beginning and end of each shift, providing the number of parts produced during that shift is less than 5,000 parts.

The samples inspected for the beginning of production and the intervals described above shall be the first five (5) parts produced for the former and the last five (5) parts produced for the latter. Parts that fail to comply with dimensional requirements after dimensional stability has been attained shall be classed defective.

If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

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(NOTE 1) - In establishing a stabilization time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g.) every 30, 60, etc. mins.) until dimensional stability is attained. The stabilization time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the stabilization time shall be sent to Picatinny Arsenal, SARPA-QA-A-S. If any new change in material, process parameters, a reworked cavity, or new cavity is used, a new stabilization time shall be established and approved.

4.3.3.2 Verification of heat treat of casing (See Dwg. 9281579), Major Defect, Code No. 07001.-Three (3) casing from each heat treat batch shall be subjected to this test. If any casing fails to comply with the requirement specified on the applicable drawing, the heat treat batch represented by the sample shall be rejected. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.3 Static burn time of fuze assembly (See Dwg. 9277712), Major Defect, Code No. 08001.-Thirty-two (32) fuze assemblies shall be selected from each lot and subjected to this test. If two (2) or more assemblies fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and in 4.3.3 (see 6.3).

4.4 Test methods and procedures.-

4.4.1 Verification of heat treat of casing.-The casing shall be subjected to the Knoop hardness reading in accordance with ASTM Test Method D1474-68. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.2 Static burn time of fuze assembly.-This test shall be conducted in accordance with a Government approved procedure.

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5. PREPARATION FOR DELIVERY

5.1 Preservatiotand packaging.-

5.1.1 Level C.-The parts shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing.-

5.2.1 Level C.-The parts, preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking.-Marking of the carton shall comply with MIL-STD-1169.

5.4 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types: Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

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6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is catalogued and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirement Lists (DD Form 1423).

The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabrication or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

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6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, Attn: SARPA-QA-A-S.

6.5 Distribution of ammunition data cards.-Distribution of data cards shall include the following: Commanding Officer, Picatinny Arsenal, Attn: SARPA-QA-A-S, Dover, New Jersey 07801.

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